

The main drawback of Bulletproofs is that they are slower to verify than both SNARKs and STARKs.

Key use cases: Monero (confidential transactions), Grin

Here's a summary of the key trade-offs between the three main ZKP types.

- **zkSNARKs:** Smallest proof size, fast verification, requires trusted setup, vulnerable to quantum attacks.
- **zkSTARKs:** Largest proof size, fastest generation, no trusted setup, postquantum secure.
- **Bulletproofs:** Medium proof size, no trusted setup, slowest verification, best for range proofs.

How ZKPs are applied in open source blockchains

Private transactions: The Zcash

example: Zcash is the most well-known example of a public blockchain using ZKPs for transaction privacy. It was built directly on the Zerocash protocol, published in 2014 by BenSasson, Chiesa, Garman, Green, Miers, Tromer, and Virza. The paper proposed using zkSNARKs to create a cryptocurrency where transactions are valid but fully private -- the sender, receiver, and amount are all hidden behind a cryptographic proof.

In Zcash, users can choose between transparent transactions (similar to Bitcoin) and shielded transactions (fully private). When a shielded transaction is made, a zkSNARK proof is generated that proves the sender has enough funds to make the payment, without revealing the actual balance. It also proves that the transaction outputs are valid (no new coins are created). The network nodes verify this proof without ever seeing the underlying data.

Zcash's Sapling upgrade in 2018 dramatically improved performance, reducing memory requirements by 97% and proof generation time by 81%, making private transactions practical on mobile devices.

Scalability through ZKRollups:

ZKPs are also being used to solve

Ethereum's scalability problem through a technology called ZKRollups. A rollup takes many transactions offchain, processes them in batches, and then uses a ZKP to prove that all transactions in the batch were processed correctly. Only the proof (and a small amount of state data) is submitted to Ethereum's main chain.

This provides two benefits at once: scalability (many transactions are compressed into one onchain proof) and privacy (the details of individual transactions are not visible on the main chain). Projects like zkSync, Polygon zkEVM, and StarkNet are leading this space.

As noted in a 2024 survey by Mohanty *et al*, ZKPs enable blockchains to shift from collective to singular computation with networkwide verification, a fundamental change in how distributed systems handle trust.

Private identity and authentication: ZKPs are also being used for privacy -- preserving identity verification on blockchains. The traditional way to prove your identity online requires you to share personal data, name, date of birth, national ID with every service you use. This creates massive privacy risks, as every service becomes a potential point of data breach.

With ZKPs, a user can prove they meet a requirement; for example, that they are over 18 years old, or that their credit score is above a certain threshold without revealing the underlying data. The proof is generated from verified credentials stored on the blockchain and can be checked by any verifier.

Zhou *et al* (2024) conducted an extensive survey of ZKPbased identity sharing systems and found that combining blockchain with ZKP technology provides both decentralised management of identity and strong privacy guarantees, addressing the vulnerability of traditional identity systems to data breaches and identity theft.

Privacy in smart contracts:

Ethereum's smart contracts are fully transparent. Anyone can see the code, the inputs, and the outputs. For many business

use cases — such as sealed bid auctions, private voting, or confidential financial agreements — this is unacceptable.

ZKPs allow smart contracts to verify that computations were done correctly without revealing the inputs. For example, in a private auction, a ZKP could prove that the winning bid is the highest one, without revealing what any of the bids actually were. The computation is done offchain, and only the proof is submitted to the smart contract.

Zhang *et al* (2024) demonstrated a data trading security scheme that combines ZKPs with smart contracts on a blockchain, showing that it is possible to verify the fairness and correctness of a data transaction without any party revealing their private data to the other.

Challenges and limitations

Despite their power, ZKPs are not a perfect solution. Several significant challenges remain.

Computational cost: Generating a ZKP is computationally intensive. Even with modern hardware, creating a zkSNARK proof can take several seconds for complex computations. While verification is fast, the cost of proof generation is still a barrier for widespread adoption, especially on mobile devices or low power systems.

Trusted setup risk (for SNARKs): As discussed, zkSNARKs require a trusted setup ceremony. If this ceremony is compromised — for example, if a participant in the MPC ceremony saves their secret instead of destroying it — the entire security of the system is at risk. An attacker with the toxic waste from the setup could create unlimited counterfeit tokens without detection.

Implementation vulnerabilities: Even when the underlying mathematics of a ZKP is sound, implementation bugs can introduce serious vulnerabilities. A 2024 analysis of ZKP systems by Tang *et al* found multiple vulnerability classes in real world ZKP implementations, including under constrained circuits, which can allow a prover to generate